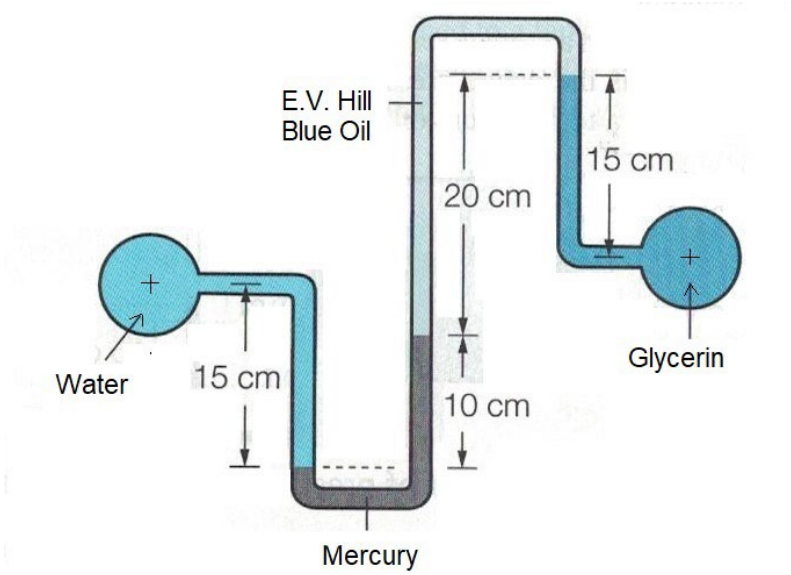
ChE 2O04 Tutorial #4

Tutorial Problem #1

For the given heights on the differential manometer, determine the pressure difference (kPa) between the two pipes that are used to transport water and glycerin. All the liquids are at room temperature (i.e. 20 °C).



Tutorial Problem #2

Consider ethanol at 20 °C flowing through a venturi meter in which the ratio of the 'throat' diameter to 'upstream' diameter is 0.5. If the absolute pressure at the 'throat' of the venturi meter is 34 psi lower than the absolute pressure immediately upstream of the venturi meter, calculate the average velocity of ethanol in the 'throat' section of the venturi meter.